

Predicting the Next Category and Region of a Visit: A Check-in-Level Multi-Task Study on Mobility Data

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Abstract—

Index Terms—next-category prediction, next-region prediction, multi-task learning, check-in-level representation, location-based social networks, mobility data

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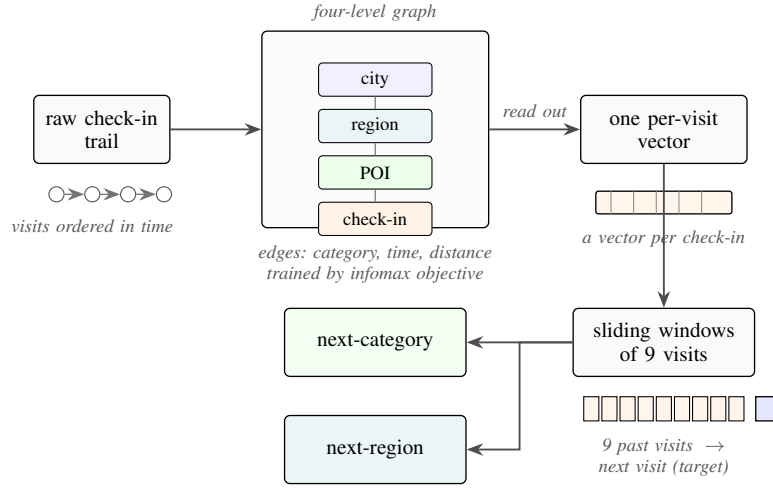


Fig. 1. From a user's check-in trail to two predictions. Each visit is placed in a four-level graph (check-in, POI, region, city) and described in its own context; overlapping nine-visit windows then feed the model, which outputs the next category and the next region.

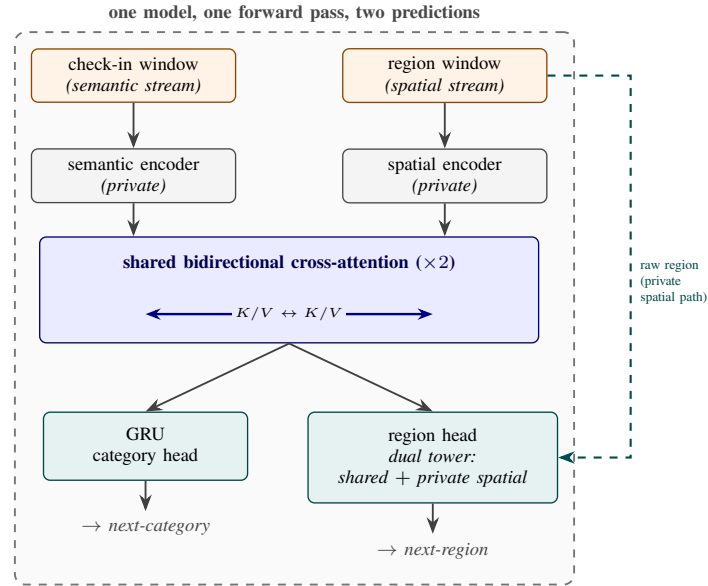


Fig. 2. The single multi-task model. Two input streams, a check-in (semantic) window and a region (spatial) window, pass through private per-task encoders into a shared bidirectional cross-attention stack (the only shared parameters). A GRU head reads the shared semantic output for next-category; a dual-tower region head combines that shared output with a private spatial tower that runs on the raw region window for next-region. One model, one forward pass, two predictions.

TABLE I

DATASET STATISTICS, ORDERED BY NUMBER OF REGIONS. FIVE GOWALLA STATES AND ONE INTERNATIONAL CITY (ISTANBUL, FROM MASSIVE-STEPS) SPAN SMALL TO LARGE REGION SPACES, CHOSEN ON PURPOSE TO TEST WHETHER THE FINDINGS HOLD ACROSS SCALES AND SETTINGS. EVERY DATASET USES THE SAME SEVEN GOWALLA ROOT CATEGORIES (COMMUNITY, ENTERTAINMENT, FOOD, NIGHTLIFE, OUTDOORS, SHOPPING, TRAVEL). WINDOWS ARE THE OVERLAPPING NINE-VISIT INPUT PLUS THE NEXT VISIT AS TARGET, UNDER THE FINAL BOARD WINDOWING (GATED STRIDE ONE, MINIMUM SEQUENCE LENGTH TEN); MAXIMUM AND AVERAGE SEQUENCE LENGTH ARE PER-USER CHECK-IN COUNTS. DENSITY IS THE CHECK-IN-MATRIX FILL, $100 \times \text{CHECK-INS}/(\text{USERS} \times \text{POIS})$, THE STANDARD NEXT-POI SPARSITY MEASURE.

Dataset	Source	Check-ins	Users	POIs	Regions	Windows	Max len	Avg len	Density (%)
Istanbul [†]	Massive-STEPS	462,615	23,694	29,816	520	270,217	817	19.5	0.065
AL	Gowalla	113,846	3,858	11,848	1,109	96,326	3,835	29.5	0.249
AZ	Gowalla	236,450	7,869	20,666	1,547	200,895	5,589	30.1	0.145
FL	Gowalla	1,407,034	21,052	76,544	4,703	1,274,418	16,679	66.8	0.087
TX	Gowalla	4,089,892	38,644	160,938	6,553	3,830,414	42,300	105.8	0.066
CA	Gowalla	3,171,380	37,090	169,145	8,501	2,925,466	14,855	85.5	0.051

Region unit: a census tract (a mahalle for Istanbul). Windows = overlapping nine-visit input plus the next visit as target, after the last-place de-skew gate (gated stride one, minimum sequence length ten). Density = $100 \times \text{check-ins}/(\text{users} \times \text{POIs})$. [†] Istanbul is the mahalle substrate (post null-coordinate drop); the raw corpus is 544,471 check-ins over 23,700 users.

TABLE II

THE CHECK-IN-LEVEL REPRESENTATION (CHECK2HGI) MAKES NEXT-CATEGORY PREDICTION FAR MORE LEARNABLE THAN A STANDARD PLACE EMBEDDING (HGI). READ THIS AS: A VECTOR PER VISIT, NOT A VECTOR PER PLACE, IS WHAT DRIVES THE CATEGORY GAIN. VALUES ARE NEXT-CATEGORY MACRO-F1 UNDER A MATCHED SINGLE-TASK HEAD, ON THE OVERLAP BOARD (ONE WINDOWING FOR THE WHOLE PAPER). THE CHECK2HGI COLUMN IS THE SINGLE-TASK CATEGORY CEILING ON THE PAPER'S SUBSTRATE; THE HGI ARM IS FILLED AT AL, AZ, FL, AND CA (A LARGE AND CONSISTENT MARGIN, +27.6 TO +39.6), WITH TX STILL BEING SCORED.

State	Check2HGI	HGI	Δ
AL	55.87	26.56	+29.31
AZ	57.13	29.50	+27.63
FL	75.15	35.53	+39.62
TX	69.95	TBD	TBD
CA	70.26	32.31	+37.95

Check2HGI = the single-task next-category ceiling on the paper's substrate (check-in level, design_k overlap engine; RESULTS_BOARD, seed 0×5 folds, fp32-matched), identical to the dedicated category column of Table III. The HGI arm is the same single-task head on the place-level HGI embedding under the identical overlap windowing; AL, AZ, FL, and CA are filled (margin +27.6 to +39.6), and TX is still being scored (TBD). A controlled test attributes most of the category gain to the per-visit context rather than to extra supervision (a per-place-averaged version of the same embedding recovers only part of it); the exact share is being re-confirmed on this substrate.

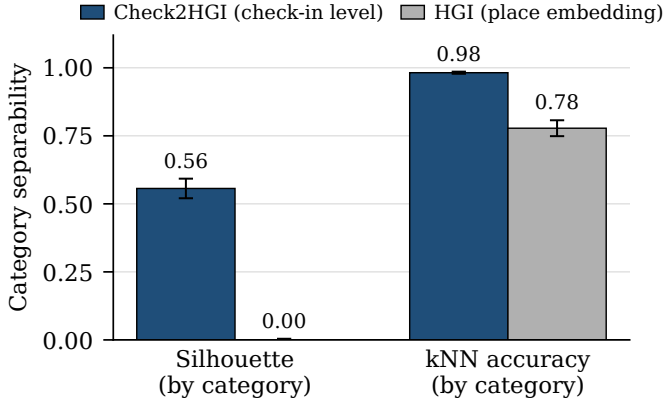


Fig. 3. Category separability of the two representations. The check-in-level representation cleanly separates the seven categories where the place embedding does not, which is exactly why it helps the category task and is neutral on region.

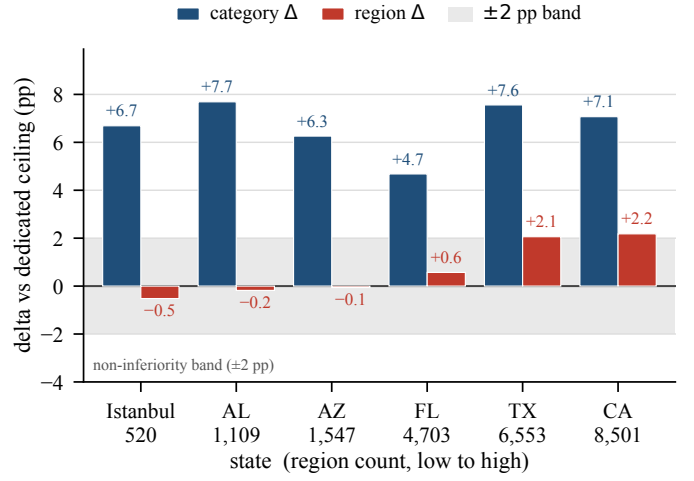


Fig. 4. Signed category and region gains of the single joint model over the dedicated ceilings, across states ordered by number of regions. The category gain is positive at every state; the region gain is positive at the large states (FL, TX, CA) and within a two-point margin at the small (AL, AZ, Istanbul), and the largest state (CA, 8,501 regions) has the largest region win. All cells are $n=5$ (seed 0) provisional.

TABLE III

ONE MODEL, TWO TASKS. READ THIS AS: A SINGLE JOINT MODEL BEATS THE DEDICATED CATEGORY CEILING AT EVERY STATE, AND ON REGION IT BEATS THE DEDICATED CEILING WHERE THE REGION SPACE IS LARGE (FL, TX, CA) WHILE MATCHING IT WITHIN A TWO-POINT MARGIN WHERE THE REGION SPACE IS SMALL (AL, AZ, ISTANBUL). THE LARGEST REGION STATE (CA, 8,501 REGIONS) HAS THE LARGEST REGION WIN, SO THE SPATIAL GAIN GROWS WITH SCALE. BOLD MARKS THE JOINT MODEL WINNING A CELL AGAINST ITS DEDICATED CEILING. FOR REGION, $\uparrow$ MARKS A BEAT (SUPERIORITY, PAIRED WILCOXON) AND $\approx$ MARKS A MATCH (STATISTICALLY NON-INFERIOR WITHIN A TWO-POINT MARGIN, TOST). CATEGORY IS MACRO-F1; REGION IS ACC@10. STATES ARE ORDERED BY NUMBER OF REGIONS. ALL JOINT-MODEL CELLS ARE $n=5$ (SEED 0) PROVISIONAL.

State	Regions	Next-category (macro-F1)				HMT-GRN [§]	ReHDM	STAN [‡]	Dedicated
		Markov	POI-RGNN	Dedicated	Joint (ours)				
Istanbul	520	24.55	30.12	53.20	59.89	60.4	–	61.86	74.80
AL	1,109	20.50	23.80	55.87	63.56	57.05	66.06	60.72	69.99
AZ	1,547	23.92	27.64	57.13	63.39	43.70	54.65	49.86	59.40
FL	4,703	29.74	34.49	75.15	79.82	63.74	65.68	– [‡]	76.71
TX	6,553	28.67	33.03	69.95	77.51	53.85	– [†]	– [†]	64.96
CA	8,501	27.58	31.78	70.26	77.33	49.61	– [†]	– [†]	63.48

[†] joint model beats the dedicated ceiling (superiority); $\approx$ joint model matches it within a two-point margin (TOST non-inferior).

Region externals, by comparability: **HMT-GRN**[§] (region-native, multi-task, end-to-end on our footing) is the **primary** comparison; STAN (Luo et al., faithful) and ReHDM are secondary references. The joint model beats the primary region-native baseline (HMT-GRN) at every state. Markov = Markov-9-cat best- K floor; POI-RGNN = next-category (macro-F1).

[‡]STAN run faithfully (its own embeddings + STAN-native prefix-expansion sequences, from raw) on our seed-0 folds / region targets, converged + audited (PR #53): A cascade multi-task comparator (CSLSL) ties our parallel joint model at equal cost (a defense, not a win; see text).

ReHDM is a published-protocol reference (its own chronological split), not a paired cell. [†] Faithful STAN and ReHDM infeasible at CA/TX scale. POI-RGNN FL/CA/TX.

[§]HMT-GRN-style (deviation ledger): its own end-to-end recurrent trunk + train-only region-transition prior on our tract targets; the graph module and hierarchical beam search.

All joint-model cells are $n=5$ (seed 0) provisional; the FL contextual-embedding control (CTLE) is reported in the text, not here.